

ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech in Electrical Engineering	Indian Institute of Technology Kanpur	9.2/10.0
2021	Class XII (ISC)	La Martiniere College, Lucknow	99.6%
2019	Class X (ICSE)	La Martiniere College, Lucknow	97.8%

SCHOLASTIC ACHIEVEMENTS

- Received the **Academic Excellence Award (twice)** for excellent performances in the academic years 2021-22 and 2022-23
- Secured **AIR 1365** in JEE Advanced among 0.15 million candidates and **AIR 1832** in JEE Main among 1.1 million candidates
- Qualified for the **KVPY 2021 (SX)** fellowship, securing **All India Rank 1680** in the examination conducted by IISc Bangalore
- Recipient of the **National Talent Search Examination** Scholarship 2019, awarded by NCERT to the **top 0.01%** candidates

RESEARCH EXPERIENCE

N-path Filters | **SURGE-IITK** | Prof. R.S. Ashwin Kumar, Dept. of Electrical Engg., IIT Kanpur (May'23 - July'23)

Objective	- To understand the working principle and study the theoretical background of N-path Filters - To evaluate the advantages of N-path filters over SAW/BAW filters in wireless receiver architectures
Approach	- Surveyed 8 relevant IEEE papers on the topic complemented with weekly discussions with the professor - Examined the implications of parasitic capacitances and switch resistance on the linearity of the filters
Results	- Compiled the recent trends and advancements made in the domain of N-path filters over the last 5 years - Analyzed methods to implement a 10 dB higher in-band linearity and a 66% reduction in input frequency

KEY PROJECTS

Retrofitting of AUV Anahita | Team AUV-IITK | Prof. Indranil Saha, Dept. of CSE, IIT Kanpur (July'22 - Oct'22)

Objective	To revamp the design of PCB Stands in the main hull and the Front Camera Casing of the vehicle Anahita
Approach	- Ensured cost and material reduction by running multiple FEA simulations on the PCB stands via Ansys Maximized the peripheral vision of the front camera by modifying its existing orientation using Solidworks
Results	Minimized the material costs by 25% and achieved a 40% increase in the front camera's field of view

Remote-Controlled Plane | Aeromodelling Club, IIT Kanpur (May'22 - July'22)

Objective	To gain hands-on experience in aeromodelling and learn aircraft fabrication and primary flight controls
Approach	- Worked in a team of 5 people to fabricate the aeroplane bodies using the Hot-Wire cutting technique - Assimilated the basics of Avionics and practiced flying, landing, and target drop on the aircraft simulator
Results	Developed and flew 2 fixed-wing aircrafts, one built of Styrofoam and the other of Balsa wood

RELEVANT COURSEWORK

Awarded A/B+ grade for exceptional performance in all of the following courses (*ongoing)

Data Structures & Algorithms-1	Data Structures & Algorithms-2*	Fundamentals of Computing
Probability and Statistics	Introduction to Electronics	Microelectronics-1
Control Systems Analysis	Signals, Systems & Networks	Partial Differential Equations

POSITION OF RESPONSIBILITY

Convener, Council of Students' Hostel Affairs | Students' Gymkhana, IIT Kanpur (March'23 - Present)

Leadership	Elected with a more than a 3/4th majority by the elected members of the Students' Senate of the institute In-charge of the smooth functioning of 15 hostels providing residence and mess facilities to 9500+ students Senator BT/BS Y21: Elected by 1200+ UG Y21 students as their representative in the Students' Senate
Operations	- Formulated Hall Allocation Policy 2023-24 and managed inter-hall shifting of 1500 post-graduate students - Ensured a smooth transition of the largest batch of 1200+ first-year students to 5 different hostels - Devised and implemented the summer vacation and convocation stay policy of 3000+ students in 7 hostels
Impact	- Executed a 12% reduction in the mess establishment charge, providing monetary relief to 4000+ students - Spearheaded a 30% crunch reduction, i.e. the largest ever, ensuring efficient utilization of the new hostel - Secured a 50% increment in the budget per hostel, totalling Rs. 14 lakh, from the Dean's Capital Fund

EXTRA-CURRICULAR ACHIEVEMENTS

Technical	Secured 1st position in Takneek'23; Built a popsicle bridge that withstood a record 88 kg load	2023
Sports	Secured 1st position out of 32 teams in the Intra-Hall Table Tennis Championship	2022
Business	Secured 2nd position out of 60 teams in the Game of Auctions competition in Entreverse	2022
Volunteering	Student Guide and Academic Mentor; Tutored 30 freshmen in the Engineering Graphics course	2022
Cultural	- Awarded the Sangeet Prabhakar certification (1st division), equivalent to graduation in Music - Secured 2nd position out of 130 teams, winning Rs.30000 in the Canara Knowledge Champs Quiz	2020 2019